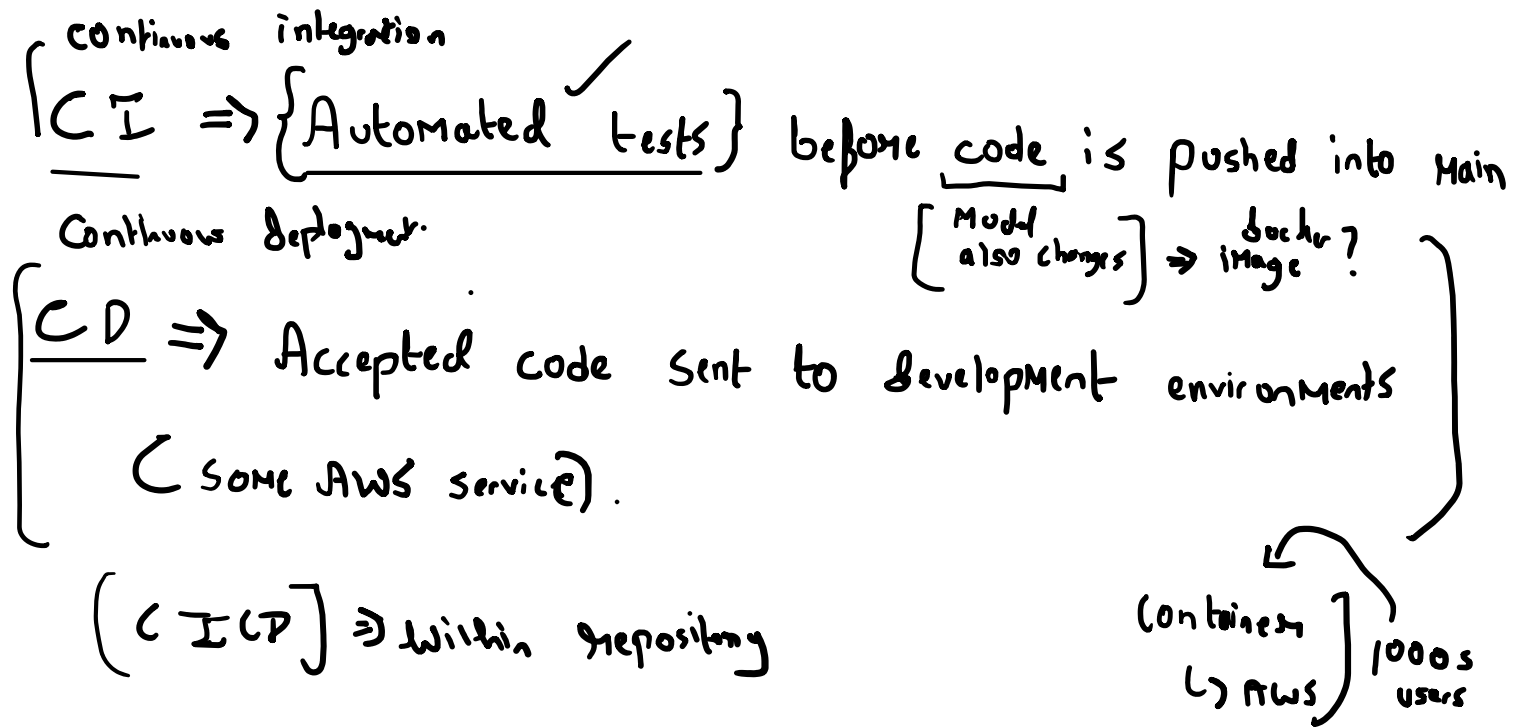




If I want to push some code to Main, tests will be run on my code before the push is accepted.

In python, pytest is used to define test cases.

feature distribution

range of Model output

Business rules

GitHub actions

GitHub Actions is a continuous integration and continuous delivery (CI/CD) platform that allows you to automate your build, test, and deployment pipeline. You can create workflows that build and test every pull request to your repository, or deploy merged pull requests to production.

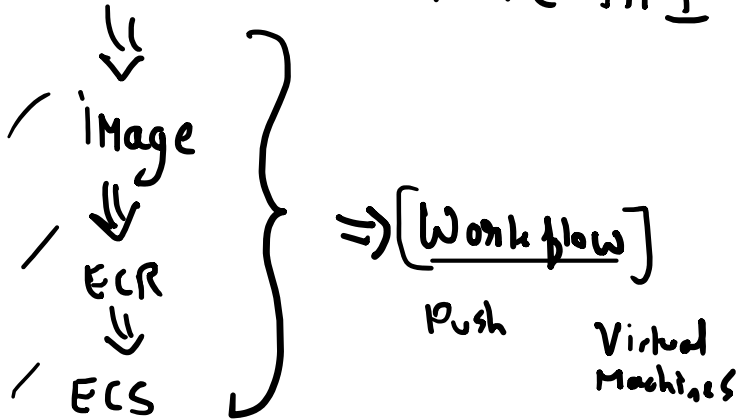
[Automate the entire process] ★

GitHub actions

Workflow file $\Rightarrow$ instructions to github

Train our Model.

Use our Model in the API



Developers
Will keep
experimenting
and pushing
code

$\Rightarrow$ Verification
Needed
(CI).

If CI checks
succeed, go
for CD

[Workflow file] $\Rightarrow$ What all needs to be checked.
in the pushed code

[Test cases
Model loading]

{Push} $\Rightarrow$ Event
or
{pull request}

$\left[\begin{array}{l} \{ \text{Push} \} \Rightarrow \text{Event} \\ \text{or} \\ \{ \text{pull request} \} \end{array} \right]$

Perform the following:

- i) Copy the new repo from developers branch.
- ii) Start a Virtual Machine for free.
Windows
Ubuntu
Mac OS
- iii) Run defined tasks.

[yaml file]

[C.I workflow file]

CI Workflow

```

name: Model Tests

on:
  push:
    branches:
      - main

jobs:
  test-model:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout repository
        uses: actions/checkout@v4
        with:
          fetch-depth: 1

      - name: Set up Python
        uses: actions/setup-python@v5
        with:
          python-version: "3.10"

      - name: Install dependencies
        run: |
          python -m pip install --upgrade pip
          pip install pytest
          pip install -r requirements.txt

      - name: Run model tests
        run: |
          python -m pytest -v
  
```

When to trigger the workflow
Defining the event

GitHub Virtual Machines

paste

git push {local} to remote

3.11

lightweight
VM

Docker image

Cd action

09 January 2026 20:09

```
name: Deploy to AWS ECS

on:
  push:
    branches:
      - main

env:
  AWS_REGION: ap-south-1 # set this to your preferred AWS region, e.g. us-west-1
  ECR_REPOSITORY: my_app # set this to your Amazon ECR repository name
  ECS_SERVICE: car-price-task-service-jbdsdme # set this to your Amazon ECS service name
  ECS_CLUSTER: car-price-cluster # set this to your Amazon ECS cluster name
  ECS_TASK_DEFINITION: task-definitions.json # set this to the path to your Amazon ECS task definition
  # file, e.g. .aws/task-definition.json
  CONTAINER_NAME: car-price-container # set this to the name of the container in the
  # containerDefinitions section of your task definition

jobs:
  job1:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v2
      - name: Install Python 3
        uses: actions/setup-python@v1
        with:
          python-version: "3.10"
      - name: Install dependencies
        run: |
          python -m pip install --upgrade pip
          pip install pytest
          pip install -r requirements.txt
      - name: Run tests with pytest
        run: pytest
```

[app will be continuously running]

```
job2:
  needs: job1
  runs-on: ubuntu-latest
  steps:
    - name: Checkout
      uses: actions/checkout@v3
    - name: Configure AWS credentials
      uses: aws-actions/configure-aws-credentials@v4
      with:
        aws-access-key-id: ${{ secrets.AWS_ACCESS_KEY_ID }}
        aws-secret-access-key: ${{ secrets.AWS_SECRET_ACCESS_KEY }}
        aws-region: ${{ env.AWS_REGION }}
    - name: Login to Amazon ECR
      id: login-ecr
      uses: aws-actions/amazon-ecr-login@v2
    - name: Build, tag, and push image to Amazon ECR
      id: build-image
      env:
        ECR_REGISTRY: ${{ steps.login-ecr.outputs.registry }}
        IMAGE_TAG: ${{ github.sha }}
      run: |
        # Build a docker container and
        # push it to ECR so that it can
        # be deployed to ECS.
        docker build -t $ECR_REGISTRY/$ECR_REPOSITORY:$IMAGE_TAG .
        docker push $ECR_REGISTRY/$ECR_REPOSITORY:$IMAGE_TAG
        echo "Image=$ECR_REGISTRY/$ECR_REPOSITORY:$IMAGE_TAG" >> $GITHUB_OUTPUT
    - name: Fill in the new image ID in the Amazon ECS task definition
      id: task-def
      uses: aws-actions/amazon-ecs-render-task-definition@07587c9d45a4930bf0e3da8dd2feb2a463cf4a3a
      with:
        task-definition: ${{ env.ECS_TASK_DEFINITION }}
        container-name: ${{ env.CONTAINER_NAME }}
        image: ${{ steps.build-image.outputs.image }}
    - name: Deploy Amazon ECS task definition
      uses: aws-actions/amazon-ecs-deploy-task-definition@v2
      with:
        task-definition: ${{ steps.task-def.outputs.task-definition }}
        service: ${{ env.ECS_SERVICE }}
        cluster: ${{ env.ECS_CLUSTER }}
        wait-for-service-stability: true
```

In GitHub Actions, what is an "event" that can trigger a workflow?

7 users have participated

A

An operational malfunction within the workflow's execution environment.

14%

B

A pre-defined time when the GitHub Actions server conducts maintenance.

14%

C

Any significant modification to GitHub's platform codebase.

0%

D

A specific activity in a repository like a pull request or push event.

71%

From <<https://www.scaler.com/instructor/meetings/i/github-actions-setting-up-cd-pipelines-25/>>